LESSON PLAN · SCRATCH (GR 6–8) INTERMEDIATE: DATA & SIMULATIONS

Lesson 2: Analysing Data: Average, Maximum, Minimum

Module 1 | 45 minutes | Data algorithms | Grades 6–8 · ages 11–14

Objective: Use traversal to calculate an average and find the maximum and minimum values in a list.

Success criteria

- I can calculate an average.
- I can find the max and min with a scan.
- I can explain why initialisation matters.

Key vocabulary

- **mean (average)**: Total divided by the number of items.
- **maximum / minimum**: The largest / smallest value.
- **linear scan**: Checking each item once, in order.

You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)

Timed lesson plan (45 min)

Time	Stage	What happens
5 min	**Warm-up (unplugged)**	Tallest in line (5 min): a “champion” card passes along a line of learners; swap when someone taller appears.
5 min	**Watch**	Play the lesson video with captions; pause at each heading and ask a question.
8 min	**Build along**	Weather report: learners follow the steps.
14 min	**Perform**	Code Lab: find the maximum, Code Lab: average
9 min	**Probe**	Debug the minimum, activity
4 min	**Wrap-up**	Exit ticket: What should max start as, and why?

Unplugged warm-up

Tallest in line (5 min): a “champion” card passes along a line of learners; swap when someone taller appears.

Teaching the idea

Mean = total/length; max/min via linear scan with a running champion initialised to the first element. Initialising to 0 is a classic bug for all-positive (min) or all-negative (max) data.

Analogy: King of the hill: the champion stays until someone stronger arrives.

Live demo: Run the min finder starting at 0 and show the wrong result.

Questions to ask

Ask	Listen for
Why start max at item 1?	It's a real value from the data.
How do you change max into min?	Swap > for <.
Why divide by length rather than 7?	Works for any list size.

Differentiation

Support
Provide the traversal loop; learners add the comparison.

Stretch
Also record WHICH day was hottest (store the index).

Common misconceptions

Misconception	How to fix it
Start max/min at 0	Start at the first item.
Average = max + min ÷ 2	That's the midrange, not the mean.

Assessment

Use the Code Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

Exit ticket
What should max start as, and why?

Homework
Find the max, min and average of something you measured this week.

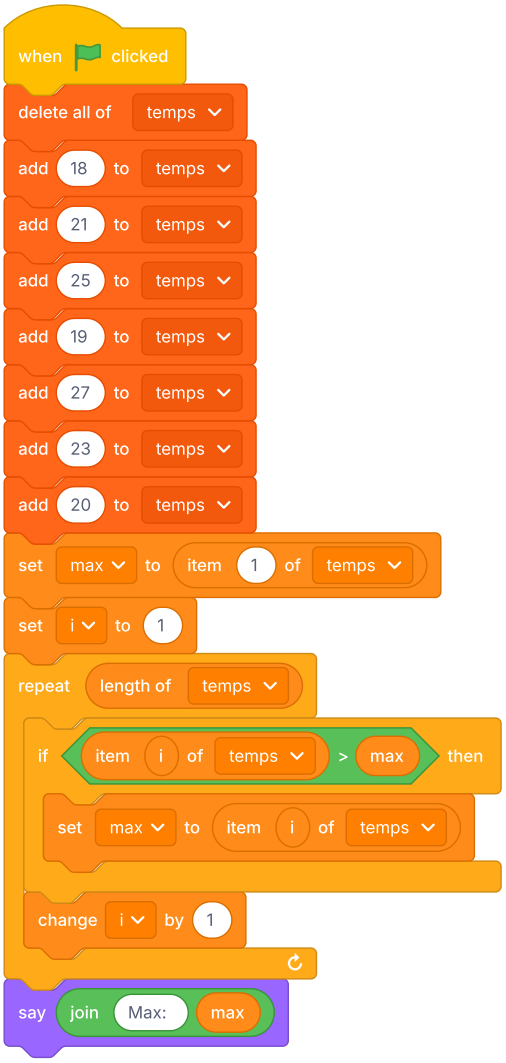
Answer key — Kitabash activities

1. Code Lab: find the maximum (perform · code lab)

Goals checked: Max is 27 · Compares in a loop · Reports it

One solution:

Analyst

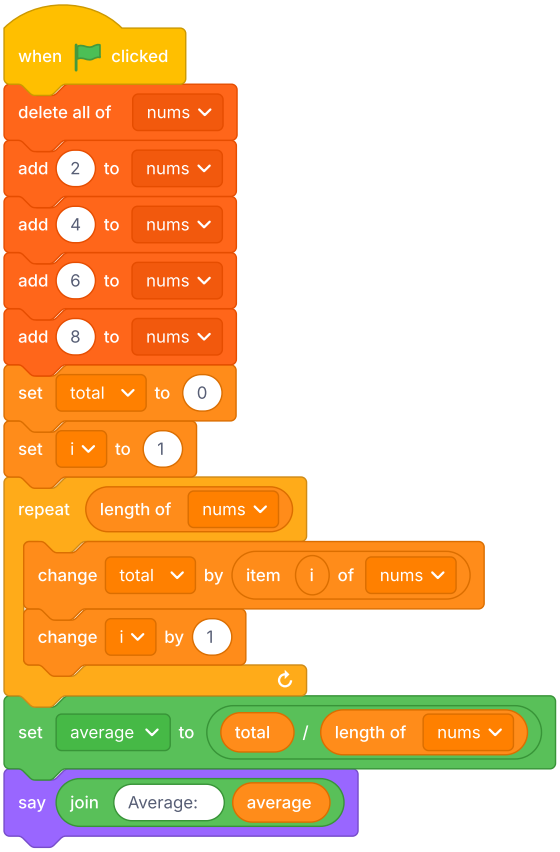


2. Code Lab: average (perform · code lab)

Goals checked: Average is 5 · Divides by the length · Calculates

One solution:

Analyst



3. Debug the minimum (probe · multiple choice)

Answer: min should start as item 1, not 0

4. Explain your thinking (probe · free response)

Look for: average, maximum, list. **Example:** The average shows how well the class did overall, so the teacher knows whether to re-teach the topic. The minimum shows if anyone is struggling a lot and might need extra help.

Answer key — worksheet

1. **Key words:** mean (average) = Total divided by the number of items. · maximum / minimum = The largest / smallest value. · linear scan = Checking each item once, in order.
2. **Plan your code:** Goals: Max is 27; Compares in a loop; Reports it.
3. **Plan your code:** Goals: Average is 5; Divides by the length; Calculates.
4. **Debug the minimum:** A — No temperature is below 0, so min never changes. Start with a real item.

5. **Explain your thinking:** The average shows how well the class did overall, so the teacher knows whether to re-teach the topic. The minimum shows if anyone is struggling a lot and might need extra help.

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